

EECS 498/CSE 598: Zero-Knowledge Proofs

Winter 2026

Lecture 15

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Agenda for this lecture

- Announcements
- ASIC-model frontends, cont'd
- Expressing programs in R1CS
- Boolean operations
- Optimizations

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Announcements

- Project 3 online, due 3/20
- Theory track problem set 1 online, due 3/20
- 598 research track: email me

• Turing award

Bennett, Brassard

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ASIC-model frontends

- Verilog / VHDL

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- **Expressing programs in R1CS**
- Boolean operations
- Optimizations

R1CS for basic program constructs

add1(X):

$y = x + 1$

$$A_Z \circ B_Z = C_Z$$

$$\begin{bmatrix} 1 & 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ x \\ y \end{bmatrix} = 0$$

$$\begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ x \\ y \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ x \\ y \end{bmatrix}$$

$$1 + x - y = 0$$

$$y = x + 1$$

R1CS for basic program constructs

$\text{add}(a, b):$

$$c = a + b$$

$$\begin{bmatrix} 0 & 1 & 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ a \\ b \\ c \end{bmatrix} = 0 \quad \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ a \\ b \\ c \end{bmatrix}$$

$$a + b - c = 0$$

$\text{mul}(a, b):$

$$c = a \cdot b$$

$$\begin{bmatrix} 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ a \\ b \\ c \end{bmatrix} = 0 \quad \begin{bmatrix} 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ a \\ b \\ c \end{bmatrix}$$

R1CS for basic program constructs

addmul(a_0, a_1, b_0, b_1):

Ret $a_0 b_0 + a_1 b_1$
 $(d = a_0 b_0 + a_1 b_1)$

→
SSA

$$c_0 = a_0 b_0$$

$$c_1 = a_1 b_1$$

$$d = c_0 + c_1$$

$[1 \ a_0 \ a_1 \ b_0 \ b_1 \ c_0 \ c_1 \ d]$

$[0 \ 1 \ \dots \]$

$[0 \ 0 \ 1 \ \dots \]$

$[\dots \ 1 \ 0]$

0 0 0 1 ...

0 0 0 0 1 ...

1 0 ... 0

0 ... 1 0 0

0 ... 0 1 0

0 ... 1

divide(a, b)

$$c = a / b$$

Conditionals in R1CS

"Theorem": Any straightline arithmetic FFP computation can use these gadgets

CondMul(X, Y, Z, b):

If $b = 1$:

Ret XY

→

$$C_1 = XY$$

$$d_0 = b C_1$$

$$b' = 1 - b$$

$$d_1 = b' C_0$$

$$C_0 = XZ$$

Else

Ret XZ

$$[1 \ X \ Y \ Z \ b \ \ b' \ C_0 \ C_1 \ d_0 \ d_1 \ d]$$

bitcheck(b)

Ret $b \in \{0, 1\}$

$$\begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ b \end{bmatrix} = \begin{bmatrix} -1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ b \end{bmatrix} = \begin{bmatrix} 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ b \end{bmatrix}$$

Conditionals in R1CS

Range Check (X)

Ret $x \in [0, 7]$

↳

$b_0, b_1, b_2 \in \{0, 1\}$

check $x = b_2 2^2 + b_1 2 + b_0$

$$\begin{bmatrix} \text{bit check} \\ \text{0-1} \end{bmatrix} \begin{bmatrix} x \\ b_0 \\ b_1 \\ b_2 \end{bmatrix} = 0$$

$[1 \dots 0]$

$[0 \dots 0]$

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R1CS for Boolean operations

Exercise: design R1CS representations for bitwise AND, OR, NOT, and XOR

R1CS for integer operations

Other operations

```
def lessthan(x, y):  
    Return (x < y)
```


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Optimizing R1CS representations